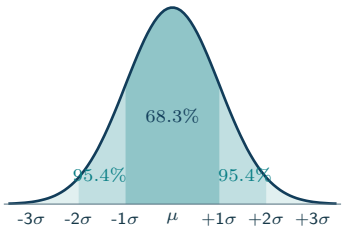
PROBABILITY LADDER 8 OF 9

The normal distribution and the central limit theorem

Sums and averages of many independent trials follow a bell curve, whatever the shape of a single trial.

1 / THE RULE AND THE PICTURE

About 68%, 95% and 99.7% lie within 1, 2 and 3 standard deviations of the mean



A normal variable with mean μ and standard deviation σ has density $\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$, and its distances from the mean are measured in the z-score $z = (x-\mu)/\sigma$:

$$P(|X-\mu| < \sigma) = 0.683, \quad P(< 2\sigma) = 0.954, \quad P(< 3\sigma) = 0.997$$

2 / USING IT / A WORKED EXAMPLE

Scores with mean 100 and standard deviation 15

Between 85 and 115 lies one standard deviation either side, so 68.3% of scores. A score above 130 is two standard deviations up, with probability $\frac{1}{2}(1-0.954) = 0.023$.
The cumulative values to remember are $\Phi(1) = 0.841$, $\Phi(2) = 0.977$, and $\Phi(1.96) = 0.975$, which gives the familiar 95% range $\mu \pm 1.96\sigma$.

3 / THE CENTRAL LIMIT THEOREM / SUMS OF DICE

As the number of dice grows, the histogram of the sum approaches the normal curve in red



A single die is flat, two dice give a triangle, and by five dice the shape is already close to the curve with the same mean $3.5n$ and variance $\frac{35}{12}n$.
The theorem says the sum of n independent copies of any variable with finite variance is approximately normal for large n.

4 / THE NORMAL APPROXIMATION TO THE BINOMIAL

$\mathrm{Bin}(100, \frac{1}{2})$ has mean 50 and standard deviation 5

To find $P(X \geq 60)$ treat X as normal and use the boundary 59.5, the continuity correction, giving $z = \frac{59.5-50}{5} = 1.9$ and a probability of 0.0287, against the exact binomial value 0.0284.
Without the half-unit correction the estimate $z = 2$ gives 0.0228, noticeably worse.

5 / AVERAGES AND THE STANDARD ERROR

The average of n readings has spread $\sigma/\sqrt{n}$

The mean of 25 rolls of a die has mean 3.5 and standard deviation $\dfrac{1.708}{\sqrt{25}} = 0.342$, so an average above 4 is $\frac{0.5}{0.342} = 1.46$ standard deviations high, with probability 0.072.
Quadrupling the sample halves the spread, which is why polls and measurements gain accuracy slowly.

ASIDE / WHY THE BELL

Adding many small independent effects blurs every distribution into the same shape, which is why heights, measurement errors and exam scores look normal.

LINKS TO OTHER SHEETS

The sums of dice start in *Random variables, expectation and variance*, and the binomial in *The binomial distribution*; the last sheet is *Information and entropy*.

Convention. Φ is the cumulative standard normal distribution; μ, σ the mean and standard deviation; probabilities from the exact normal and binomial formulas.